

Energy Consumption of DECT-2020 NR Mesh Networks

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Abstract—ETSI DECT-2020 New Radio (NR) is a new flexible radio interface targeted to support a broad range of wireless Internet of Things (IoT) applications. Recent reports have shown that DECT-2020 NR achieves good delay performance and it has been shown to fulfill both massive machine-type communications (mMTC) and ultra-reliable low latency communications (URLLC) requirements for 5th generation (5G) networks. A unique aspect of DECT-2020 as a 5G technology is that it is an autonomous wireless mesh network (WMN) protocol where the devices construct and uphold the network independently without the need for base stations or core network architecture. Instead, DECT-2020 NR relies on part of the network devices taking the role of a router to relay data through the network. This makes deployment of a DECT-2020 NR network affordable and extremely easy, but due to the nature of the medium access protocol, the routing responsibility adds an additional energy consumption burden to the nodes, who in the IoT domain are likely to be equipped with a limited battery capacity. In this paper, we analyze by system level simulations the energy consumption of DECT-2020 NR networks with different network sizes and topologies and how the reported low latencies can be upheld given the energy constraints of IoT devices.

Keywords—IoT, DECT-2020, 5G, NR, mesh network, energy consumption

I. INTRODUCTION

The DECT-2020 NR standard was released by the European Telecommunications Standards Institute (ETSI) in 2020. It is intended for industrial Internet of Things (IoT) applications requiring either massive machine-type (mMTC) and/or ultra-reliable low latency (URLLC) communications.

DECT-2020 NR introduces a device-based wireless cluster-tree mesh network, capable of forming an autonomous network without the need for separate base stations or core network. Nodes can act as routers between their neighbors, forwarding data between them in case direct connection is not possible.

With wireless mesh networking (WMN) the coverage can theoretically reach extreme distances with minimal costs and effortless deployment. However, forwarding data through the network with a reasonable delay requires that each node passing the data onwards has an adequately active duty cycle, meaning the node can receive and transmit reasonably often so that the delay requirements are met, and the data does not accumulate in relaying nodes. However, maintaining an active duty cycle requires energy resources which may be scarce if the network consists of battery-operated nodes, as is common in the IoT domain.

Being a new technology, DECT-2020 NR has not been extensively studied yet. In [1] the authors introduced the DECT-2020 NR standard and evaluated its performance by comparing it to the existing low-power wide-area network (LPWAN) solutions. The paper demonstrated delay performance similar to existing solutions but with much higher node densities. In [2] the standard was shown to fulfill the International Telecommunication Union Radiocommunication Sector (ITU-R) requirements for 5G networks, but in [3] only a partial fulfillment of these requirements was confirmed. These studies paid only little consideration to the energy required to reach the presented level of performance. Only in [1] the issue was touched by stating that a multi-hop network was more energy efficient than a single-hop network when considering the amount of energy needed for transmissions. We will demonstrate in this paper that this viewpoint is one-sided since the energy consumption of a multi-hop mesh network is not limited to the transmissions.

Energy consumption of WMNs is a well-known problem covered by multiple studies in recent years [5], [6]. Like [1], most of them consider the energy consumption primarily as a routing problem, indicating that the energy is mostly consumed by the device transmissions. However, unlike many earlier WMN technologies, DECT-2020 NR uses state-of-the-art physical layer techniques, such as Cyclic Prefix Orthogonal Frequency Division Multiplexing (CP-OFDM), turbo coding and Hybrid ARQ (HARQ) with incremental redundancy which decrease the duration and increase the energy efficiency of the transmissions, making the comparison to old studies difficult and thus justifying the need for separate investigation on DECT-2020 NR.

This paper extends the WMN energy problem field of study by exploring how the advanced features of DECT-2020 NR affect the WMN energy consumption, whether different network topologies have an impact on this and also whether the recently reported low latency capabilities of the new standard can be upheld in typical IoT scenarios given the energy constraints of IoT devices.

II. RESEARCH METHODOLOGY

Computer simulations are a valuable tool to study the behavior of complex networks or systems when the alternatives, analytical evaluation or building a real-life test environment, are difficult or impossible to use. This is the case for a newly released technology standard such as

DECT-2020 NR for which no devices nor networks yet exist.

For this study, we used an open-source Network Simulator 3 platform [7] into which we implemented a module for DECT-2020 NR. The module consists of the MAC and PHY layers implemented according to ETSI TS 103 636 series specifications [8]. In the following, the system simulator modeling details are explained.

A. Network organization

The DECT-2020 NR standard explains how the network organizes itself. For a single network, there is always one sink, i.e. a gateway node to the Internet. There can be multiple sinks in a group of nodes, but from the system point of view, the sinks form their own networks (cluster trees), which can co-exist, but operate independently from each other. In this study, we concentrate on the operation of a single network entity under one sink.

Organization starts by the sink selecting FT (Fixed Termination point) mode. In FT mode, a node first performs background scanning, searching for the least busy channel, out of the available DECT channels, for operation. After finding a channel, the FT starts beaconing on it.

All other nodes are in PT (Portable Termination point) mode, where they initially scan channels for beacons and after detecting one or more of them, select from them the sender FT for association. A node having multiple FT candidates makes the choice based on the route cost. Exact route cost calculation is left to implementation, but essentially the node is free to use any information available. In this study, the nodes select the strongest beaconing FT for association.

After a PT has successfully associated with an FT, the PT switches to FT mode and performs operations as explained for the FT mode above, in order to invite potential children and by that, extend the coverage area of the network.

Naturally, not all nodes end up associated with child nodes. These nodes are left as pure PT nodes, i.e. leaf nodes. After the network organization is finished, these nodes stop beaconing and go to sleep, only to wake up listening to their parent's beacon or to transmit data.

The organization cycle continues until all nodes within a coverage area are associated with another node and the network is complete.

B. Network topology

The finished network can be thought of as nested node tiers, each tier adding a node-to-node hop for the traffic traveling towards the sink.

Depending on the network characteristics, such as node density, scenario size, beacon transmission power and required beacon quality for association, there can be only one or multiple tiers in the network, forming either a star or a mesh topology.

1) *Star topology:* A single-tier network where all nodes are within sink coverage and associate directly with it. All nodes apart from the sink are leaf nodes and do not have any other role. A simple example of a star topology is shown in Fig. 1.

2) *Mesh topology:* Nodes out of sink coverage associate to other nodes in FT mode, who become routers. In addition to acting as a child PT to its own parent FT, a router acts as a parent to its own child node(s). A router node provides a route to the sink for the data of its PTs. Fig. 2 shows a simple example of a multi-tier mesh network.

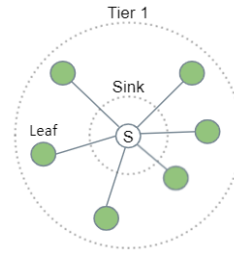


Fig. 1 Simple example of a star topology with a sink and 6 leaf nodes.

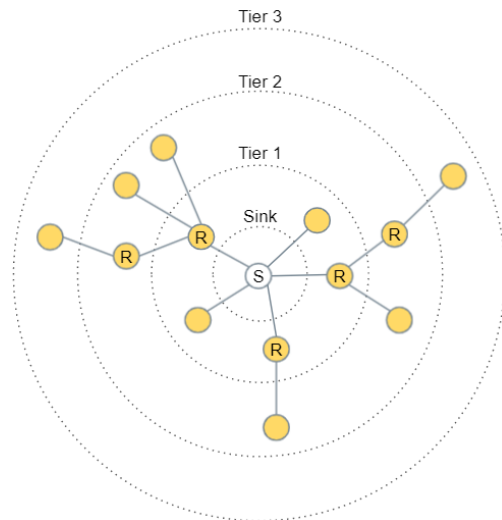


Fig. 2 Simple example of a 4-tier mesh topology consisting of a sink, 5 routers and 8 leaf nodes.

C. Data transmission

After network organization, data transmission starts. Fig. 3 shows a simple example diagram of data being transmitted from a leaf to the sink via a router. The operation of leaf nodes is very efficient: they are only awake to listen to their parent's beacon and to transmit data. Other times they can sleep. In most IoT use cases, nodes generate a small amount of data relatively infrequently. Thus, it is expected that leaf nodes can be asleep for most of the time.

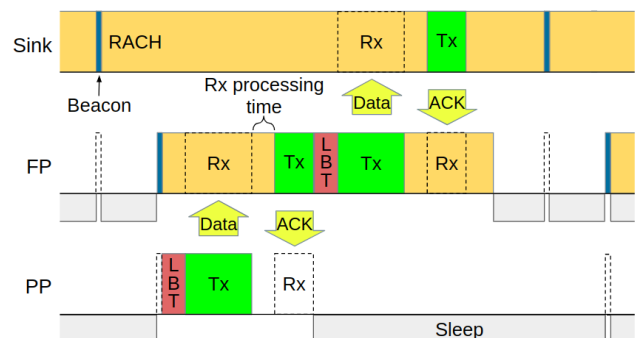


Fig. 3 Simple example of data forwarding from a leaf to the sink via a router.

A router on the other hand, needs to keep its receiver on periodically for the duration of its random access channel (RACH) period in order to be able to receive data from its children. This increases the activity of the router and decreases its energy efficiency.

Sink is the gateway node to the Internet, so it is assumed to be powered by a wall outlet or a Power over Ethernet (PoE) connection and thus its energy usage is practically irrelevant. Without power limitations, it can offer tier 1 nodes continuous RACH.

Data traffic from each node is targeted to the sink and the nodes transmit data to their parent's RACH. The PHY transmission is preceded by a Listen-Before-Talk (LBT) and is subject to HARQ until it is decoded successfully or dropped due to reaching the maximum number of retransmissions. If data is dropped, it is lost as there is no upper layer ARQ.

III. SIMULATIONS

We simulated a DECT-2020 NR network in a reasonably large IoT scenario consisting of 1000 nodes. The most central node was selected as the sink. Scenario radius was varied in order to capture the effects of both star and mesh topologies with different amounts of node tiers. Table I shows general simulation parameters and assumptions.

TABLE I. SIMULATION PARAMETERS

Parameter	Value
Scenario radius	400 m - 5 km
Frequency	1.89 GHz
Available DECT channels	10
Propagation model	ITU Urban Macro
Shadowing	Disabled
Channel model	5G spatial channel model [10]
Number of nodes	1000
Number of sinks	1
Antenna model	1x1 omni, 0 dBi gain
Noise figure	7 dB
Beacon Tx power	23 dBm
Min acceptable beacon quality for association	3 dB
Power control	Pathloss based: $TxP = \min(23 \text{ dBm}, -68 \text{ dBm} - 0.7 * PG)$, where PG is path gain
Max no of HARQ retransmissions	9
Max HARQ processes per node	2
SCS scaling factor	1
FFT scaling factor	1
LBT min CW	8
LBT max CW	64

We considered two different IoT use cases: high and low traffic intensity, with tight and loose delay requirements, respectively. Table II shows the simulated use cases and their parametrization. A data packet payload of 100 B was an approximation of an average amount required for typical one-time IoT remote sensing [9].

In the simulations, we measured the end-to-end packet delay, battery life expectation and the service rate, i.e. the percentage of the data that is successfully received by the target node. The end-to-end delay was measured from the time a packet was generated to the time when it was successfully received at the target node.

TABLE II. SIMULATION USE CASES

	High traffic intensity	Low traffic intensity
Real-life use case examples	Smart home condition monitoring, Smart transport quality conditions monitoring, Smart environment tsunami / avalanche / earthquake / landslide detection [9]	Smart energy demand response management (DRM), Smart transport road condition monitoring, Smart agriculture irrigation / fertilization / pest control [9]
Tolerable delay	Few seconds	Few minutes
Beacon interval	32 s	32 s
Router RACH period per beacon period	64 x [100, 200, 400] ms	1 x [5, 10, 20] ms
Traffic	100 B data packet per 60 s per node	100 B data packet per 24 h per node
Traffic direction	Unidirectional, from sensing node to the sink	

The router duty cycle is determined by the RACH period, and is a key factor in determining the achievable delay and also the battery life. Strict delay requirements require an active duty cycle, whereas looser requirements allow a less active one. For both use cases, we simulated a range of RACH periods in order to find the best option.

For high traffic intensity, we set multiple short RACH periods per beacon period. This minimizes the data waiting time while allowing the router to sleep intermittently and save energy.

Node battery life expectation was based on experienced modem energy consumption during the simulation as the radio is assumed to be the main energy consumer in the IoT device. Energy is consumed during the simulation according to the electric current used in the states the nodes are in. State transitions did not consume any additional energy.

Table III shows the used energy parameters. The used currents in different states were based on actual values of an existing Nordic Semiconductor nRF9160 LTE-M/NB-IoT modem [12] and it is expected that the values reflect a realistic range of an upcoming DECT-2020 NR modem also.

TABLE III. ENERGY PARAMETERS

Parameter	Value
Battery capacity	18 kJ (5 Wh)
Voltage	3.7 V
"Sleep" state current	8 μ A [12]
"Rx" or "LBT" state current	45 mA [12]
"Tx" state current	Linear Tx current model [11], where current (A) = TxP (W) / $(3.7 \text{ V} * 0.37) + 0.045 \text{ A}$

We analyzed the energy consumption of all nodes, apart from the sink, during the stable network state. Thus, the consumption during the network organization phase was not accounted for in the battery life estimation.

IV. SIMULATION RESULTS

Fig. 4 shows the percentage of nodes per tier with different scenario radii. The number of network tiers increases with scenario radius. A network has star topology only if the nodes are within 400 m from the sink with the used assumptions. Wider coverage area can only be achieved via mesh. The larger the scenario size, the more tiers the network needs in order to cover all nodes in the scenario.

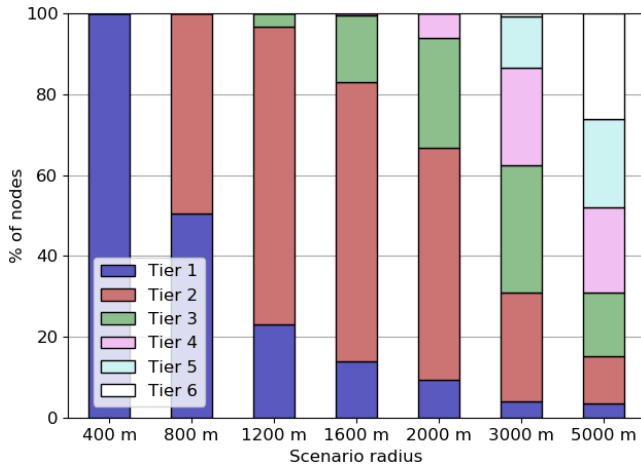


Fig. 4 Percentage of nodes in different network tiers with different scenario radii.

Fig. 5 shows examples of simulated network topologies with selected scenario radii. The figure demonstrates how the network is a star topology when nodes are within a small radius, but as the nodes are dispersed over a larger area, a multi-tier cluster tree network starts to form.

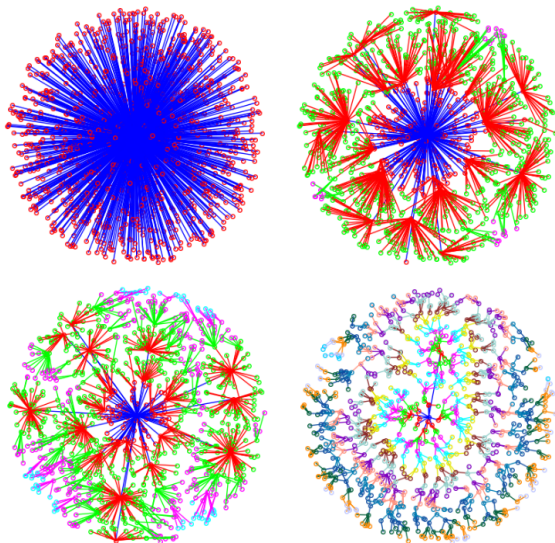


Fig. 5 Examples of simulated network topologies with 400 m (top left), 1200 m (top right), 2000 m (bottom left) and 5000 m (bottom right) scenario radii.

A. High traffic intensity results

From end-to-end packet delays, shown in Fig. 6, we notice that each simulated case seems to be able to serve the traffic within an acceptable delay. Delay increases as the mesh radius is increased and at 5 km the delay is at the limits of delay tolerance with the lowest duty cycle. Increasing duty cycle activity has a decreasing impact on delay.

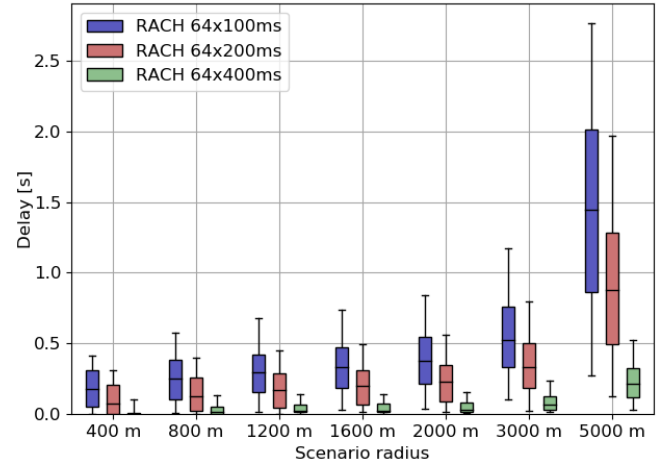


Fig. 6 Box plots showing 5th, 25th, 50th, 75th and 95th percentile of end-to-end delay for the high intensity traffic use case with different RACH settings and scenario radii.

However, even though we are within delay tolerance, there are data throughput issues with low duty cycles. Fig. 7 shows the service rate of simulated cases. It can be seen that with lower duty cycles, a significant amount of data is not served, i.e. data is dropped at some point en route to the sink.

With higher duty cycles, the service rate is near 100% so an active duty cycle is clearly needed in order to ensure service reliability.

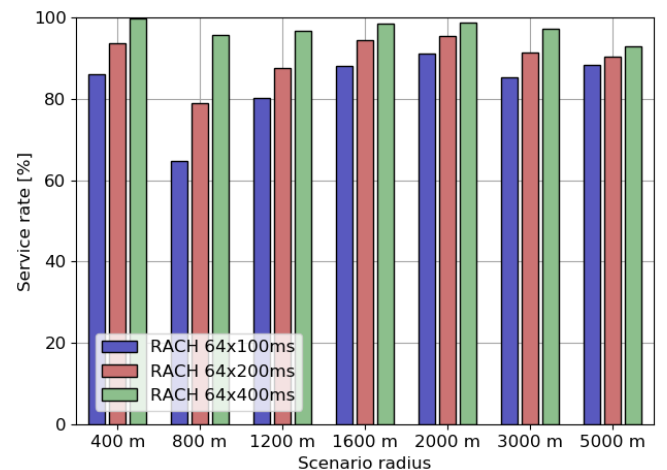


Fig. 7 Network service rate for the high intensity traffic use case with different RACH settings and with different scenario radii.

The reason for the service rate decrease with lower duty cycles is the fact that data builds up in the routers. Thus, the routers have more data to transmit during RACH occasions, which decreases the success rate of those transmissions due to longer packets having a higher probability of collisions.

This is visible in Fig. 8 from which we can also see that the longer the scenario radius, the lower is the collision probability. This is because the number of children per router decreases as the number of routers increase.

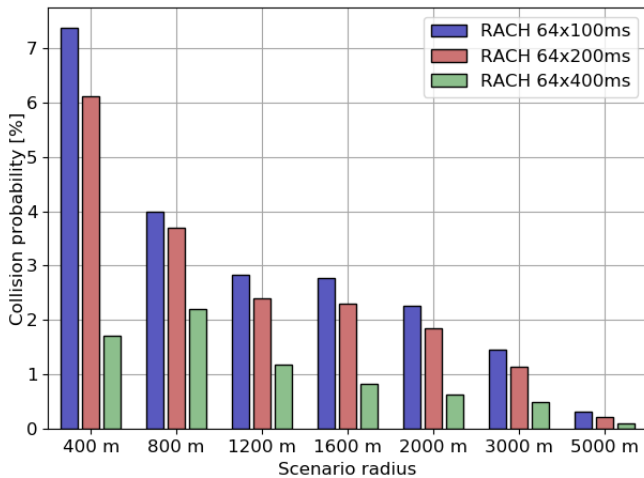


Fig. 8 Collision probability for the high intensity traffic use case with different RACH settings and scenario radiuses.

Average battery life of router nodes is shown in Fig. 9 with different RACH settings and scenario radiuses. Router lifetimes are measured in days with the given assumptions. The 400 m radius case is omitted since the scenario is a star topology without routers.

The figure demonstrates the significance of the duty cycle to the router battery life. Lower duty cycles seem to offer reasonable delays, but as we saw earlier, this comes from the somewhat compromised reliability of the service. On the other hand, using higher duty cycles offers a low delay and a reliable service but leads to a very limited router battery life.

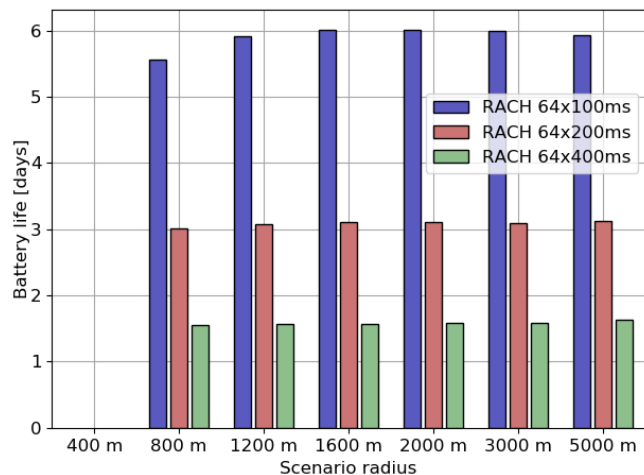


Fig. 9 Average battery life of router nodes for the high intensity traffic use case with different RACH settings and scenario radiuses.

Correspondingly, Fig. 10 shows the lifetimes of leaf nodes, which are measured in years as opposed to days in the routers. In leaf nodes, the differences come mainly from the transmission distances, which increase as the radius increases and nodes are more sparsely located in the scenario.

With the 400 m radius star topology, the transmission distances are longer than in the 800 m radius mesh topology

and thus leaf node lifetimes are shorter in average. Still, in the total network lifetime, a star topology totally surpasses a mesh network, and with a moderate number of nodes, it is clearly a better option. However, increasing the number of children of a single parent quickly increases competition of the medium, leading to an increased number of collisions and thus worse service.

If the network has to cover a large area, it needs to form a mesh topology. If the latency requirements are also tight, routers need an active duty cycle. This combination limits the lifetime of a battery-operated network.

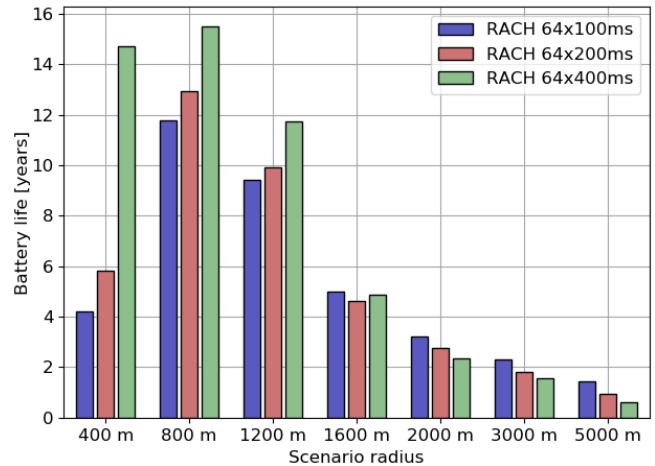


Fig. 10 Average battery life of leaf nodes for the high intensity traffic use case with different RACH settings and scenario radiuses.

B. Low traffic intensity results

With the low traffic intensity use case, the requirements for delay are looser and thus it is possible to parametrize the network to minimize the energy consumption. Thus, the router duty cycle is decreased, but should still be set high enough so that the data does not accumulate in the transmitter buffers.

For the selected traffic intensity of one 100 B packet per 24 hours, we used a 5, 10, and 20 ms RACH period occurring once per beacon interval.

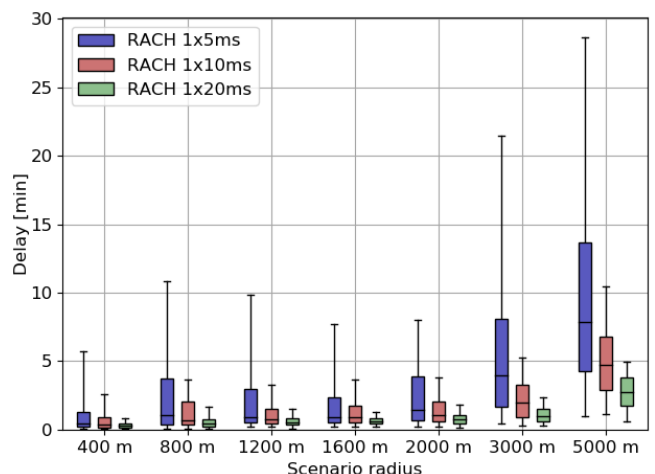


Fig. 11 Box plots showing 5th, 25th, 50th, 75th and 95th percentile of end-to-end delay for the low intensity traffic use case with different RACH settings and scenario radiuses.

Fig. 11 shows the end-to-end delays. With the lowest duty cycle, the delays can reach tens of minutes. For this

particular use case, long delays are not necessarily a problem as long as the traffic is reliably delivered to the target.

However, from the service rates in Fig. 12, it can be seen that the lowest duty cycle cannot fully serve the offered traffic. Only with 10 and 20 ms RACH period occurring once per 32 s does the service rate reach above 90 % in all network radiuses.

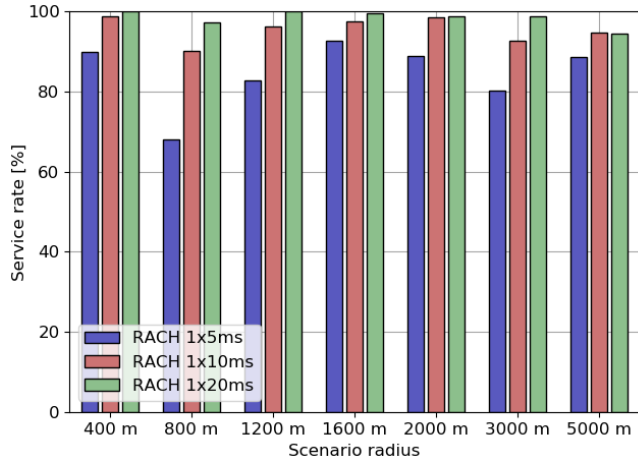


Fig. 12 Network service rate for the low intensity traffic use case with different RACH settings and scenario radiuses.

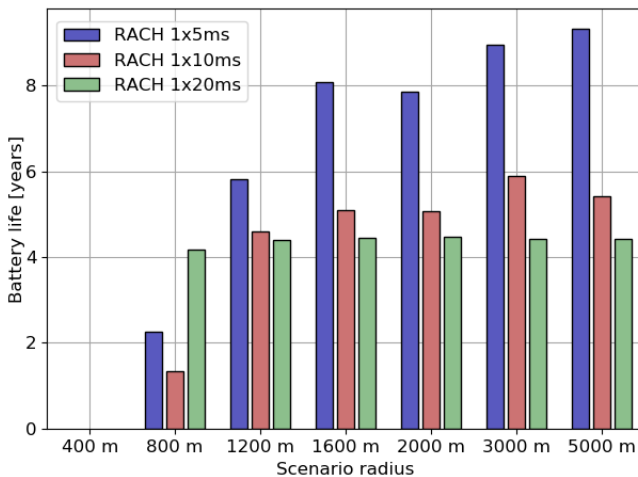


Fig. 13 Average battery life of router nodes for the low intensity traffic use case with different RACH settings and scenario radiuses.

Fig. 13 shows the expected battery life of mesh routers with different scenario radiuses. With the low intensity traffic use case, the duty cycle again has a significant impact. Small changes to duty cycles have tremendous effects on router battery lives. With the given assumptions, an average router lasts for years, which is an adequate lifetime for the underlying real-life IoT use cases [9].

V. CONCLUSIONS

A system level simulation analysis of DECT-2020 NR confirms its good delay performance with some notable constraints. A wide-range battery-operated mesh network cannot maintain both long uptime and low latency. It was noticed that rather than the actual transmissions, maintaining an active duty cycle to ensure low latencies is the most important consumer of energy in a DECT-2020 NR mesh network. In order to guarantee a reasonable network uptime, the duty cycle of battery-operated mesh routers and, consequently, the requirements for traffic delay should be low. For delay sensitive data, the alternative is to either use mains powered routers or to use multiple sinks, covering the area solely with star topologies, without the need for the network to resort to mesh mode at all.

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